



Aim To Top Review and Training Center - Iligan
Agricultural and Biosystems Engineering Licensure Examination
Review Session

MASTERING AGRICULTURAL PROCESSING

100+ CONCEPT-BASED QUESTIONS FOR RICE AND CORN

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INTRODUCTION

Agricultural and Bio-Processing refers to local activity or series of activities to maintain or raise the quality or change the form or characteristics of agricultural, fishery, forestry and biological products/materials and includes, but is not limited to, cleaning, sorting, grading, treating, drying, dehydrating, grinding, mixing, milling, canning, dressing, slaughtering, freezing, pasteurizing, conditioning, packaging, repacking, transporting of agricultural, fishery, forestry and other biological products/materials

RA 10915

QUESTION 1

What is the main goal of primary processing in agricultural products?

- a. To increase nutritional value
- b. To convert into ready-to-eat forms
- c. To prepare raw materials for further processing
- d. To improve taste and aroma

ANSWER



c. To prepare raw materials for further processing

Explanation: Primary processing prepares raw materials (e.g., shelling, cleaning, drying) for secondary processing. Others describe objectives of secondary or value-adding processes.

QUESTION 2

Which of the following is a primary processing activity?

- a. Shelling of corn
- b. Baking of bread
- c. Vacuum packaging
- d. Juice extraction

ANSWER



a. Shelling of corn

Explanation: Shelling is a primary step to separate kernels from the cob. Others are secondary or value-added processing activities.

QUESTION 3

Which of the following activities best represents secondary processing?

- a. Sorting fruits by size
- b. Drying palay
- c. Milling corn into flour
- d. Cleaning harvested vegetables

ANSWER



c. Milling corn into flour

Explanation: Secondary processing transforms raw materials into consumable or usable forms. Milling is transformation, while others are preparatory steps.

QUESTION 4

Which type of hazard is most associated with chemical residues in vegetables?

- a. Physical
- b. Chemical
- c. Biological
- d. Ergonomic

ANSWER



b. Chemical

Explanation: Chemical hazards include pesticides and cleaning agents. Physical = foreign objects, Biological = microbes.

QUESTION 5

In food safety, what does GMP primarily aim to control?

- a. Nutrient loss
- b. Hygienic practices and production conditions
- c. Mechanical damage
- d. Market price

ANSWER



b. Hygienic practices and production conditions

Explanation: GMP ensures cleanliness and proper handling to prevent contamination. It is a basic requirement for food safety.

QUESTION 6

Which of the following is NOT part of Good Manufacturing Practices (GMP)?

- a. Wearing hairnets and gloves
- b. Washing hands before food handling
- c. Adding preservatives to extend shelf life
- d. Cleaning food contact surfaces regularly

ANSWER



c. Adding preservatives to extend shelf life

Explanation: Adding preservatives is a formulation decision, not a GMP requirement. The rest are part of hygienic practices.

QUESTION 7

A mango drying facility in Pangasinan must register with which agency to comply with local food safety regulation?

- a. DENR
- b. DTI
- c. FDA Philippines
- d. PhilRice

ANSWER



c. FDA Philippines

Explanation: The Food and Drug Administration (FDA) regulates processed food facilities. Others have different mandates.

QUESTION 8

Which is most critical in achieving uniform drying of palay in solar drying?

- a. Frequent turning or stirring
- b. Covering with plastic
- c. Using a weighing scale
- d. Spraying water before drying

ANSWER



a. Frequent turning or stirring

Explanation: Turning ensures even exposure to sunlight and air. Other options don't directly ensure uniform drying.

QUESTION 9

HACCP focuses on identifying:

- a. Nutrient levels
- b. Microorganism counts
- c. Critical control points for food safety
- d. Packaging options

ANSWER



c. Critical control points for food safety

Explanation: HACCP identifies points where hazards can be prevented or reduced. It's a proactive food safety system.

QUESTION 10

In a HACCP plan, which of the following is a Critical Control Point (CCP) in milk pasteurization?

- a. Storage room temperature
- b. Pasteurization temperature and time
- c. Milk color
- d. Cleaning schedule

ANSWER



b. Pasteurization temperature and time

Explanation: Pasteurization time/temp is vital for food safety. Others are important but not CCPs for microbial control.

QUESTION 11

A worker notices foreign materials in dried fish during packaging. What GMP principle applies?

- a. Traceability
- b. Physical hazard prevention
- c. HACCP monitoring
- d. Nutritional labeling

ANSWER



b. Physical hazard prevention

Explanation: Detecting physical hazards like bones, stones, or plastic is a key food safety and GMP concern.

QUESTION 12

If a cooperative in Bukidnon is packaging vacuum-sealed coffee beans, what processing level is this?

- a. Primary
- b. Secondary
- c. Postharvest
- d. Tertiary

ANSWER



b. Secondary

Explanation: Vacuum-sealing is a secondary processing or packaging step. Postharvest is broader and includes both.

QUESTION 13

Which is the first step in implementing HACCP?

- a. Conduct hazard analysis
- b. Establish CCPs
- c. Establish record-keeping
- d. Determine critical limits

ANSWER



a. Conduct hazard analysis

Explanation: You must first identify potential hazards before defining CCPs and limits.

QUESTION 14

What is the purpose of Critical Limits in HACCP?

- a. Set measurable safety thresholds
- b. Provide cleaning procedures
- c. Determine selling price
- d. Establish shelf life

ANSWER



a. Set measurable safety thresholds

Explanation: Critical limits define acceptable levels for time, temperature, pH, etc., to control hazards.

QUESTION 15

Which Philippine law primarily governs food safety from farm to table?

- a. RA 10068
- b. RA 10611
- c. RA 10918
- d. RA 7586

ANSWER



b. RA 10611

Explanation: RA 10611 is the Food Safety Act of 2013. RA 10068 is Organic Agriculture Act.

QUESTION 16

Which of the following is a biological hazard in food?

- a. Glass fragments
- b. Detergent residue
- c. Salmonella bacteria
- d. Excess salt

ANSWER



c. Salmonella bacteria

Explanation: Biological hazards include bacteria, viruses, and parasites. The rest are physical/chemical.

QUESTION 17

Which is the best example of secondary processing in Philippine root crops?

- a. Drying cassava slices
- b. Making cassava chips
- c. Harvesting cassava
- d. Washing and brushing roots

ANSWER



b. Making cassava chips

Explanation: Making chips involves cutting, seasoning, and frying, secondary transformation steps.

QUESTION 18

Which is most essential for preventing contamination in food processing workers?

- a. Wearing hairnet
- b. Handwashing with soap
- c. Taking a bath
- d. High salary

ANSWER



b. Handwashing with soap

Explanation: Handwashing is the most basic and effective hygiene practice.

QUESTION 19

A food processor must monitor metal contamination in processed nuts. This is part of:

- a. Sensory evaluation
- b. Physical hazard control
- c. Marketing
- d. Milling efficiency

ANSWER



b. Physical hazard control

Explanation: Metal shavings or parts are physical hazards that must be prevented or detected.

QUESTION 20

Which processing method improves shelf life by removing water at low temperature and pressure?

- a. Solar drying
- b. Oven drying
- c. Freeze drying
- d. Boiling

ANSWER



c. Freeze drying

Explanation: Freeze drying retains nutrients and improves shelf life by sublimating water in frozen form.

QUESTION 21

A banana chip processor in Davao is required to display expiration dates on packaging. This is part of:

- a. Labeling standards under food safety laws
- b. GMP layout requirement
- c. HACCP analysis
- d. Shelf-life extension

ANSWER



a. Labeling standards under food safety laws

Explanation: Labeling requirements ensure traceability and consumer protection.

QUESTION 22

Which is the most appropriate packaging material for vacuum-sealed dried mangoes?

- a. Woven sack
- b. Laminated plastic with foil
- c. Paper bag
- d. Mesh net

ANSWER



b. Laminated plastic with foil

Explanation: Laminated plastic with foil provides moisture barrier and strength for vacuum sealing.

QUESTION 23

What hazard is likely if raw and cooked products are stored together?

- a. Cross-contamination
- b. Physical breakage
- c. Color fading
- d. Loss of flavor

ANSWER



a. Cross-contamination

Explanation: Pathogens from raw foods can contaminate cooked ones without proper separation.

QUESTION 24

Which of the following does not directly improve food safety but improves product marketability?

- a. Good packaging
- b. Expiration date
- c. Food coloring
- d. Cleaning facilities

ANSWER



c. Food coloring

Explanation: Food coloring can improve appearance but does not contribute to safety.

QUESTION 25

A local coconut vinegar processor adds correct labels, includes net weight, and lists ingredients. This practice promotes:

- a. Product transparency and consumer safety
- b. Higher acidity
- c. Longer fermentation
- d. Cheaper production

ANSWER



a. Product transparency and consumer safety

Explanation: Proper labeling is a key food safety and consumer protection practice.

QUESTION 26

During inspection of a corn mill, inspectors note that product buildup is found on internal surfaces of a grinder. Which GMP principle is most violated?

- a. Personnel hygiene
- b. Sanitary design and maintenance of equipment
- c. Proper labeling
- d. Food traceability

ANSWER



b. Sanitary design and maintenance of equipment

Explanation: Buildup indicates failure in cleaning and maintenance, a core principle of sanitary equipment design. Others don't apply.

QUESTION 27

A HACCP audit team identifies that temperature logs were falsified during pasteurization of milk. What critical HACCP principle was breached?

- a. Determining critical limits
- b. Verification
- c. Record-keeping and documentation
- d. Hazard analysis

ANSWER



c. Record-keeping and documentation

Explanation: Accurate records are essential for food safety verification. Falsifying logs undermines the entire HACCP system.

QUESTION 28

What is the most suitable critical limit for drying corn intended for long-term storage?

- a. 20% MC
- b. 18% MC
- c. 16% MC
- d. 14% MC

ANSWER



d. 14% MC

Explanation: 14% moisture content (MC) is the safe limit for long-term grain storage. Higher values encourage spoilage/mold.

QUESTION 29

In a vacuum-packaged longganisa production line, which hazard is most likely if cold chain is broken?

- a. Anaerobic bacterial growth
- b. Presence of eggshell fragments
- c. Loss of spice aroma
- d. Increased salt content

ANSWER



a. Anaerobic bacterial growth

Explanation: Anaerobic bacteria thrive in oxygen-limited environments. Without refrigeration, pathogens like *C. botulinum* may grow.

QUESTION 30

What is the main purpose of verification in a HACCP system?

- a. To visually inspect workers' hygiene
- b. To confirm that the HACCP system is functioning effectively
- c. To eliminate all food hazards
- d. To clean equipment

ANSWER



b. To confirm that the HACCP system is functioning effectively

Explanation: Verification checks whether the food safety system works through audits, sampling, and testing.

QUESTION 31

A slaughterhouse is required to follow “zero contact” between carcasses and walls/floor during dressing. This is based on which principle?

- a. Sanitation Standard Operating Procedure (SSOP)
- b. Traceability
- c. Contamination prevention by design
- d. Food labeling

ANSWER



c. Contamination prevention by design

Explanation: Preventing contact avoids cross-contamination.
It's a hygienic design rule applied in slaughter operations.

QUESTION 32

If a HACCP plan lacks a clear corrective action when a limit is exceeded, what is compromised?

- a. Control of food safety risk
- b. Physical processing speed
- c. Market competitiveness
- d. Product labeling accuracy

ANSWER



a. Control of food safety risk

Explanation: Without corrective actions, you can't ensure product safety after a deviation.

QUESTION 33

In a turmeric powder facility in Nueva Ecija, what hazard is most critical to monitor during grinding and packaging?

- a. Dust particle inhalation and contamination
- b. Nutrient degradation
- c. pH increase
- d. Overheating of packaging machine

ANSWER



a. Dust particle inhalation and contamination

Explanation: Fine powder can pose health and contamination hazards; proper ventilation and filtering are essential.

QUESTION 34

Which aspect of GMP ensures that a dried mango processor documents who handled and packed each batch?

- a. Sanitary design
- b. Traceability and record-keeping
- c. Personnel hygiene
- d. Calibration

ANSWER



b. Traceability and record-keeping

Explanation: Traceability ensures accountability from source to end user, vital during recalls.

QUESTION 35

A HACCP team decides that vacuum-sealing cooked meat eliminates oxygen and microbial risk. This is:

- a. Validation
- b. Cleaning procedure
- c. Hazard analysis and CCP determination
- d. Product development

ANSWER



c. Hazard analysis and CCP determination

Explanation: Identifying and analyzing microbial risk leads to setting CCPs like vacuum-sealing.

QUESTION 36

An equipment calibration log shows irregular verification of temperature sensors in a drying chamber. What consequence is most critical?

- a. Inaccurate drying, leading to unsafe moisture levels
- b. Slower processing time
- c. Uneven coloring of product
- d. Increased energy use

ANSWER



a. Inaccurate drying, leading to unsafe moisture levels

Explanation: Without correct sensor data, moisture may not reach safe storage levels, posing microbial risk.

QUESTION 37

Which combination of hazard categories best represents GMP concerns?

- a. Biodegradability, shelf-life, and odor
- b. Nutrient content, vitamin loss, and price
- c. Physical, chemical, and biological contaminants
- d. Airflow, solar intensity, and ambient temperature

ANSWER



c. Physical, chemical, and biological contaminants

Explanation: GMP addresses all three major hazard types that could compromise food safety.

QUESTION 38

In food processing, which activity is considered a prerequisite program (PRP) rather than a CCP?

- a. Regular sanitation of surfaces
- b. Metal detector at packaging line
- c. Pasteurization time monitoring
- d. pH testing of pickled vegetables

ANSWER



a. Regular sanitation of surfaces

Explanation: PRPs include general sanitation and good practices. CCPs are specific hazard-control steps.

QUESTION 39

During high ambient temperatures in Central Luzon, what food safety risk increases in sun drying of fish?

- a. Lipid oxidation
- b. Mechanical damage
- c. Microbial spoilage due to prolonged drying
- d. Salt crusting

ANSWER



c. Microbial spoilage due to prolonged drying

Explanation: Without proper control, rapid spoilage or contamination may occur from slow, uneven drying.

QUESTION 40

In designing a plant layout for a small-scale pineapple canning plant, which principle should most influence flow design?

- a. Cost of materials
- b. Linear product flow with no cross-contamination
- c. Height of ceiling
- d. Packaging/ canning materials

ANSWER



b. Linear product flow with no cross-contamination

Explanation: Flow design must avoid raw and finished product crossing paths to reduce contamination.

QUESTION 41

What is the primary goal of a Corrective Action in HACCP?

- a. To penalize employees
- b. To regain control and prevent unsafe food from reaching consumers
- c. To lower production cost
- d. To repeat the hazard analysis

ANSWER



b. To regain control and prevent unsafe food from reaching consumers

Explanation: Corrective actions ensure deviations are addressed and products are held or reprocessed safely.

QUESTION 42

If a GMP audit finds poor lighting in inspection areas, what risk is associated?

- a. Biological contamination
- b. Missed detection of defects or foreign matter
- c. Physical contamination
- d. Overpacking

ANSWER



b. Missed detection of defects or foreign matter

Explanation: Poor lighting leads to oversight in quality control and foreign object identification.

QUESTION 43

What documentation is not optional under GMP and HACCP systems?

- a. Cleaning records
- b. Consumer preference survey
- c. Staff leave schedule
- d. Supplier marketing flyer

ANSWER



a. Cleaning records

Explanation: Cleaning logs help verify hygiene and trace contamination during audits.

QUESTION 44

Which Philippine agency provides training and support on food safety for MSMEs in agrifood processing?

- a. Bureau of Customs
- b. DOH
- c. DOST
- d. DA-BAR

ANSWER



c. DOST

Explanation: DOST-FNRI and DOST-ITDI offer technical support for MSMEs on safety and food processing.

QUESTION 45

A farmer in Isabela wants to process turmeric into powder. What is the most critical post-drying step for quality and safety?

- a. Drying
- b. Packaging
- c. Proper grinding and sieving
- d. Marketing

ANSWER



c. Proper grinding and sieving

Explanation: Finer, consistent grind reduces microbial hiding places and ensures product uniformity.

QUESTION 46

Which of the following is a preventive control measure rather than a corrective one?

- a. Disposing rejected batches
- b. Calibrating temperature sensors
- c. Recalling unsafe products
- d. Increasing preservatives

ANSWER



b. Calibrating temperature sensors

Explanation: Calibration is a preventive action to ensure safety before a deviation occurs.

QUESTION 47

In a wet market, chicken is often exposed without refrigeration. This violates which food safety concern?

- a. Packaging
- b. Cross-contamination
- c. Time-temperature control
- d. Labeling

ANSWER



c. Time-temperature control

Explanation: Meat must be kept at specific cold temps to prevent bacterial growth.

QUESTION 48

What is the main food safety concern in open-air fermentation of bagoong (fermented fish paste)?

- a. Inconsistent temperature and microbial activity
- b. Unattractive appearance
- c. Longer shelf life
- d. Darker color

ANSWER



a. Inconsistent temperature and microbial activity

Explanation: Fermentation must be controlled to prevent pathogenic overgrowth.

QUESTION 49

Why must food contact surfaces be made of stainless steel in food processing?

- a. Cheaper than aluminum
- b. Non-reactive, easy to sanitize, and corrosion-resistant
- c. Makes food taste better
- d. Keeps food hot

ANSWER



b. Non-reactive, easy to sanitize, and corrosion-resistant

Explanation: Stainless steel is ideal for food-grade surfaces due to its durability and sanitation potential.

QUESTION 50

In HACCP, what step ensures that pasteurization equipment consistently hits the required temperature?

- a. Product inspection
- b. Hazard analysis
- c. Monitoring
- d. Distribution

ANSWER



c. Monitoring

Explanation: Monitoring ensures critical parameters (like time and temp) are maintained at each CCP.

QUESTION 51

Which postharvest activity provides the most immediate improvement in the shelf life of freshly harvested root crops?

- a. Sorting by size and shape
- b. Curing under controlled humidity
- c. Proper drying to reduce surface moisture
- d. Grading based on maturity

ANSWER



c. Proper drying to reduce surface moisture

Explanation: While sorting and curing are important, drying directly reduces microbial risk by limiting water activity.

QUESTION 52

In cassava processing, what operation best prevents cyanide poisoning from raw roots?

- a. Peeling
- b. Drying under the sun
- c. Soaking followed by fermentation
- d. Grating

ANSWER



c. Soaking followed by fermentation

Explanation: Fermentation breaks down cyanogenic compounds. Drying or grating alone may not fully detoxify cassava.

QUESTION 53

Among the following, which best describes the transformation involved in secondary processing of rice?

- a. Harvesting mature panicles
- b. Shelling grains
- c. Milling brown rice into white rice
- d. Drying palay to 14% MC

ANSWER



c. Milling brown rice into white rice

Explanation: Milling changes the form and edibility of rice, a key characteristic of secondary processing.

QUESTION 54

Which processing technology would best add value to raw calamansi in a small-scale Philippine enterprise?

- a. Washing and sorting
- b. Extracting juice and converting into concentrate
- c. Air drying the fruits
- d. Packing fresh fruits in baskets

ANSWER



b. Extracting juice and converting into concentrate

Explanation: Juice concentration significantly adds value and shelf life; others are basic pre-processing.

QUESTION 55

What is the most efficient method for drying turmeric while preserving color and bioactive compounds?

- a. Open-air drying on concrete
- b. Smoke drying
- c. Cabinet drying at controlled temperature
- d. Sun drying on woven mats

ANSWER



c. Cabinet drying at controlled temperature

Explanation: Cabinet drying maintains consistent low temperatures and prevents pigment degradation.

QUESTION 56

Which of the following is the most critical in achieving high milling recovery in rice?

- a. High temperature drying
- b. Manual harvesting
- c. Uniform and gentle drying
- d. Rough handling of sacks

ANSWER



c. Uniform and gentle drying

Explanation: Rapid or uneven drying causes stress cracks, reducing whole grain yield during milling.

QUESTION 57

In the production of banana chips, which step represents a shift from primary to secondary processing?

- a. Peeling and slicing
- b. Drying in sunlight
- c. Frying and packaging
- d. Harvesting mature bananas

ANSWER



c. Frying and packaging

Explanation: Frying and packaging convert raw bananas into a finished, shelf-stable product, key features of secondary processing.

QUESTION 58

A cooperative processes ginger into candy. What step has the most influence on final texture?

- a. Cleaning and peeling
- b. Sugar concentration during boiling
- c. Packaging
- d. Air drying

ANSWER



b. Sugar concentration during boiling

Explanation: Sugar content affects crystallization and chewiness. The others have minor effects on texture.

QUESTION 59

Which primary processing technique improves both marketability and storability of garlic?

- a. Sorting by size
- b. Proper curing in well-ventilated area
- c. Washing to remove dirt
- d. Weighing and labeling

ANSWER



b. Proper curing in well-ventilated area

Explanation: Curing allows outer layers to dry and harden, which reduces rot and improves shelf life and market appeal.

QUESTION 60

What is the key reason why fluidized bed drying is used for maize and similar grains in commercial settings?

- a. Low capital cost
- b. Uniform heat transfer and rapid drying
- c. Lower labor requirement
- d. No need for moisture monitoring

ANSWER



b. Uniform heat transfer and rapid drying

Explanation: Fluidized beds ensure faster, even drying, especially for grains prone to case hardening.

QUESTION 61

Which of the following improves drying efficiency and energy savings in mechanical dryers?

- a. Increasing air humidity
- b. Reducing airflow speed
- c. Pre-cleaning grains before drying
- d. Stacking grains in thick layers

ANSWER



c. Pre-cleaning grains before drying

Explanation: Pre-cleaning prevents airflow blockage and improves heat transfer, reducing energy usage.

QUESTION 62

In tomato paste production, what secondary processing step most affects product consistency?

- a. Peeling
- b. Packaging
- c. Evaporation or concentration step
- d. Sterilization

ANSWER



c. Evaporation or concentration step

Explanation: The concentration phase reduces water and defines viscosity, critical for paste quality.

QUESTION 63

Which of the following steps is most essential to reduce microbial contamination in dried mango production?

- a. Sorting mangoes
- b. Packing in plastic
- c. Proper blanching before drying
- d. Transporting in refrigerated trucks

ANSWER



c. Proper blanching before drying

Explanation: Blanching kills surface microbes, setting a safe baseline before dehydration.

QUESTION 64

Among the following, which best improves starch quality in corn-based snack production?

- a. High moisture content
- b. Sun drying on plastic sheets
- c. Proper gelatinization through precise heat application
- d. Overcooking during frying

ANSWER



c. Proper gelatinization through precise heat application

Explanation: Gelatinization changes starch properties, impacting texture and digestibility.

QUESTION 65

Which combination of practices is ideal in increasing shelf life and quality of vacuum-packed roasted peanuts?

- a. Roasting, coloring, and sun drying
- b. Peeling and adding preservatives
- c. Roasting at correct temperature and moisture sealing
- d. Using strong spices to mask spoilage

ANSWER



c. Roasting at correct temperature and moisture sealing

Explanation: Moisture removal plus sealing reduces microbial and oxidative spoilage.

QUESTION 66

Which is the most efficient packaging option for brown rice that retains freshness during tropical storage?

- a. Woven plastic sacks
- b. Paper bags
- c. Multi-layer vacuum pouch with oxygen barrier
- d. Metal cans

ANSWER



c. Multi-layer vacuum pouch with oxygen barrier

Explanation: Brown rice has high oil content. Barrier packaging prevents oxidation and insect entry.

QUESTION 67

A calamansi processor wants to minimize vitamin C loss. Which step is most important?

- a. Labeling
- b. Heat treatment and storage condition control
- c. Juice clarification
- d. Color enhancement

ANSWER



b. Heat treatment and storage condition control

Explanation: Vitamin C is heat- and light-sensitive; controlling these is crucial for nutrient retention.

QUESTION 68

Which equipment yields more uniform flour particle size from dried cassava chips?

- a. Hammer mill
- b. Pin mill
- c. Stone grinder
- d. Rice husker

ANSWER



b. Pin mill

Explanation: Pin mills offer finer, more uniform particle sizes ideal for flour production.

QUESTION 69

In fish drying, which step most prevents spoilage during unpredictable Philippine rainy seasons?

- a. Sorting
- b. Manual turning
- c. Use of solar tunnel or hybrid dryer
- d. Covering fish with plastic

ANSWER



c. Use of solar tunnel or hybrid dryer

Explanation: Hybrid dryers offer controlled drying independent of weather fluctuations.

QUESTION 70

In cacao fermentation, which factor most influences final flavor profile?

- a. Initial size of beans
- b. Duration and temperature of fermentation
- c. Final drying temperature
- d. Packaging material

ANSWER



b. Duration and temperature of fermentation

Explanation: Flavor precursors are formed during fermentation. Duration and heat are critical.

QUESTION 71

Why are some fruits blanched before freezing?

- a. To improve color
- b. To inactivate enzymes that cause browning
- c. To increase water content
- d. To add flavor

ANSWER



b. To inactivate enzymes that cause browning

Explanation: Blanching stops enzymatic activity, helping preserve texture and color in frozen fruits.

QUESTION 72

A peanut processor applies oil-roasting instead of dry-roasting. What is the primary benefit?

- a. Stronger peanut aroma
- b. Easier packaging
- c. More even heat transfer and uniform cooking
- d. Longer frying time

ANSWER



c. More even heat transfer and uniform cooking

Explanation: Oil allows better heat distribution, leading to a consistent roast.

QUESTION 73

Which combination ensures best quality in freshly milled corn grits?

- a. Hand shelling and sun drying
- b. Mechanical shelling, uniform drying, and calibrated milling
- c. Dehusking and sorting
- d. Manual grading and slow drying

ANSWER



b. Mechanical shelling, uniform drying, and calibrated milling

Explanation: Mechanized and uniform steps ensure consistent output and product safety.

QUESTION 74

For powdered turmeric intended for export, what processing upgrade is most beneficial?

- a. Switching to sun drying
- b. Using stainless steel grinder and vacuum packaging
- c. Selling in local public markets
- d. Sorting by root size

ANSWER



b. Using stainless steel grinder and vacuum packaging

Explanation: Hygiene and quality assurance are critical for export-grade turmeric.

QUESTION 75

In processing coconut sugar, which step has most influence on final taste and texture?

- a. Filtering sap
- b. Controlled heating and stirring
- c. Drying granules
- d. Packaging in jars

ANSWER



b. Controlled heating and stirring

Explanation: Heating caramelizes sugars and defines flavor, texture, and color of coco sugar.

QUESTION 76

What is the main value-added product of processed coconut sap in Quezon and Laguna provinces?

- a. Coconut oil
- b. Coconut sugar
- c. Desiccated coconut
- d. Copra

ANSWER



b. Coconut sugar

Explanation: Coconut sugar is a high-value product derived from sap (not kernel) and increasingly exported.

QUESTION 77

The process of removing latex from freshly harvested mangoes using water or lime is called:

- a. Leaching
- b. Brining
- c. De-sapping
- d. Soaking

ANSWER



c. De-sapping

Explanation: De-sapping is done to remove latex and avoid black sap stains before fresh fruit packaging.

QUESTION 78

In cacao processing, which postharvest step most directly affects chocolate flavor development?

- a. Drying
- b. Fermentation
- c. Roasting
- d. De-shelling

ANSWER



b. Fermentation

Explanation: Fermentation initiates biochemical reactions that define flavor precursors in cacao beans.

QUESTION 79

What is the traditional Philippine product made from boiling coconut milk until oil separates and solids caramelize?

- a. Latik
- b. Copra
- c. Bukayo
- d. Panutsa

ANSWER



a. Latik

Explanation: Latik is both the residue and oil product used in Filipino desserts, made by boiling coconut milk.

QUESTION 80

Which high-value crop is typically converted into wine and vinegar in Ilocos and Cagayan Valley regions?

- a. Papaya
- b. Sugarcane
- c. Banana
- d. Coconut

ANSWER



b. Sugarcane

Explanation: Basi is a local sugarcane wine fermented and processed traditionally in Ilocos.

QUESTION 81

In pineapple processing, what is the main by-product used for animal feed or fiber extraction?

- a. Pineapple juice
- b. Peel residue
- c. Crown and stem waste
- d. Dried pulp

ANSWER



c. Crown and stem waste

Explanation: Crown and stem can be used for fiber or as livestock feed. Juice and pulp are main products.

QUESTION 82

Which term refers to the movement of goods from farm to final consumer, including logistics and warehousing?

- a. Marketing chain
- b. Supply chain
- c. Value chain
- d. Postharvest route

ANSWER



b. Supply chain

Explanation: The supply chain includes the entire physical and logistical flow of goods and services.

QUESTION 83

In value chain analysis, which actor is typically responsible for bulk transport and volume buying from farmers?

- a. Input suppliers
- b. Consolidators or middlemen
- c. Retailers
- d. Exporters

ANSWER



b. Consolidators or middlemen

Explanation: Consolidators connect farmers and processors by aggregating produce for scale economies.

QUESTION 84

What is the goal of “cold chain” systems in high-value crops like mango and lettuce?

- a. Prolong flavor
- b. Maintain quality and reduce spoilage
- c. Reduce labor
- d. Maximize drying rate

ANSWER



b. Maintain quality and reduce spoilage

Explanation: Cold chain systems ensure temperature control from harvest to consumption, extending shelf life.

QUESTION 85

What is the processing term for removal of unwanted particles from coffee beans before roasting?

- a. Milling
- b. Winnowing
- c. De-hulling
- d. Washing

ANSWER



b. Winnowing

Explanation: Winnowing separates light impurities like husks from heavier beans, common in coffee postharvest.

QUESTION 86

In mango puree production, which process separates fibrous pulp from juice?

- a. Filtering
- b. Centrifuging
- c. Pulping
- d. Mixing

ANSWER



c. Pulping

Explanation: Pulping separates coarse fibers from the juice, producing smoother puree.

QUESTION 87

What local product results from drying and powdering chili or siling labuyo?

- a. Chili sauce
- b. Chili powder
- c. Siling mantika
- d. Pickled sili

ANSWER



b. Chili powder

Explanation: Chili powder is a processed form of dried chili, used as seasoning or spice.

QUESTION 88

In value chain mapping, which part adds the most economic value to raw mangoes?

- a. Harvesting
- b. Sorting and grading
- c. Processing into dried mango or juice
- d. Transport to local market

ANSWER



c. Processing into dried mango or juice

Explanation: Processing into shelf-stable, exportable forms significantly increases product value.

QUESTION 89

What Philippine processed product is made by fermenting glutinous rice with coconut milk and sugar, then drying it?

- a. Kalamay
- b. Suman
- c. Puto
- d. Biko

ANSWER



a. Kalamay

Explanation: Kalamay is a sweet, sticky product with long shelf life, traditionally made from glutinous rice and coconut.

QUESTION 90

In supply chain terms, what is the role of "wholesalers"?

- a. Provide farm inputs
- b. Deliver to households
- c. Distribute in bulk to retailers
- d. Manage export papers

ANSWER



c. Distribute in bulk to retailers

Explanation: Wholesalers buy in large quantities from processors or consolidators and sell to retail channels.

QUESTION 91

The process of gradually removing moisture at low heat to preserve bioactives in malunggay leaves is called:

- a. Blanching
- b. Vacuum frying
- c. Air or tray drying
- d. Roasting

ANSWER



c. Air or tray drying

Explanation: Air or tray drying is suitable for heat-sensitive herbs like malunggay to preserve nutrients.

QUESTION 92

Which high-value product is derived from coconut kernel that has been dried and grated?

- a. Coconut milk
- b. Desiccated coconut
- c. Coconut sap
- d. Coconut cream

ANSWER



b. Desiccated coconut

Explanation: Desiccated coconut is made by drying grated coconut meat, commonly used in baking and exports.

QUESTION 93

What terminology best defines a chain of activities from raw material acquisition to final delivery of product to customer?

- a. Logistics chain
- b. Value chain
- c. Marketing strategy
- d. Enterprise plan

ANSWER



b. Value chain

Explanation: Value chain includes not just physical movement, but all activities that add value to the product.

QUESTION 94

What type of processing converts fresh ginger into a ready-to-use candy, powder, or juice?

- a. Secondary processing
- b. Pre-processing
- c. Cleaning
- d. Transportation

ANSWER



a. Secondary processing

Explanation: These forms change physical and chemical properties, defining characteristics of secondary processing.

QUESTION 95

In calamansi juice processing, which step is essential to prevent browning during storage?

- a. Pasteurization
- b. Addition of citric acid or antioxidants
- c. Mechanical juicing
- d. Sealing in PET bottles

ANSWER



b. Addition of citric acid or antioxidants

Explanation: Antioxidants prevent enzymatic browning and off-flavor development in citrus-based products.

QUESTION 96

Which of the following terms refers to adding extra steps to raw produce to enhance market value?

- a. Loss reduction
- b. Value addition
- c. Mechanization
- d. Sorting

ANSWER



b. Value addition

Explanation: Value addition increases product price and shelf life by transforming raw products.

QUESTION 97

Which government agency in the Philippines helps MSMEs develop value-added products through packaging and labeling support?

- a. NFA
- b. BSWM
- c. DOST
- d. PCA

ANSWER



c. DOST

Explanation: DOST through its SETUP program assists small processors with product development and labeling.

QUESTION 98

In the context of agri-food systems, what term refers to changing the form of raw products to meet specific market demands?

- a. Postharvest
- b. Processing
- c. Marketing
- d. Supply

ANSWER



b. Processing

Explanation: Processing adds value by altering the form or stability of raw agricultural products.

QUESTION 99

Which of the following best describes the goal of value chain development in agri-processing?

- a. Increase farm gate prices
- b. Reduce postharvest loss
- c. Improve competitiveness and returns across all actors
- d. Encourage organic farming

ANSWER



c. Improve competitiveness and returns across all actors

Explanation: A value chain approach seeks equitable improvements across the chain, farmers, processors, and traders alike.

QUESTION 100

Which of the following processed mango products from the Philippines has the highest demand in export markets?

- a. Fresh sliced mangoes
- b. Mango vinegar
- c. Dried mangoes
- d. Mango puree in cans

ANSWER



c. Dried mangoes

Explanation: Dried mangoes are the most widely exported and recognized processed mango product from the Philippines due to their long shelf life, convenience, and consistent demand in markets like the U.S., Japan, and South Korea.

QUESTION 101

The natural process by which fruits continue to respire and ripen after harvest is called:

- a. Ethylene burst
- b. Transpiration
- c. Climacteric rise
- d. Senescence

ANSWER



c. Climacteric rise

Explanation: The climacteric rise refers to the sudden increase in respiration rate associated with ripening in climacteric fruits like mango and banana.

QUESTION 102

What postharvest term refers to the loss of water from harvested produce, leading to weight loss and shriveling?

- a. Respiration
- b. Transpiration
- c. Dehydration
- d. Wilting

ANSWER



b. Transpiration

Explanation: Transpiration is the evaporation of water from plant surfaces. It's a key postharvest concern as it leads to reduced market quality.

QUESTION 103

Which compound is known as the ripening hormone in many fruits?

- a. CO_2
- b. O_2
- c. Ethylene
- d. Acetylene

ANSWER



c. Ethylene

Explanation: Ethylene is a plant hormone that promotes ripening in climacteric fruits such as tomato, banana, and mango.

QUESTION 104

The final stage of postharvest life when a product deteriorates beyond edibility is called:

- a. Senescence
- b. Respiration
- c. Maturation
- d. Softening

ANSWER



a. Senescence

Explanation: Senescence is the natural aging process that leads to irreversible deterioration and death of plant tissues.

QUESTION 105

In postharvest handling, the term “curing” refers to:

- a. Fermenting fruits before packing
- b. Allowing outer layers to dry and heal wounds
- c. Washing to remove latex
- d. Refrigerating freshly harvested produce

ANSWER



b. Allowing outer layers to dry and heal wounds

Explanation: Curing is a technique used for crops like onions, garlic, and sweet potatoes to reduce microbial infection and moisture loss during storage.

QUESTION 106

The temperature below which chilling injury symptoms begin to appear in tropical fruits is known as:

- a. Minimum growth temp
- b. Critical chilling temperature
- c. Freezing point
- d. Storage setpoint

ANSWER



b. Critical chilling temperature

Explanation: Each tropical crop has a specific chilling sensitivity. Exposure to suboptimal cold temperatures can cause discoloration and pitting.

QUESTION 107

A technique where produce is dipped into hot water to kill surface pathogens is known as:

- a. Hot air treatment
- b. Steam washing
- c. Hot water treatment
- d. Blanching

ANSWER



c. Hot water treatment

Explanation: Hot water treatment (HWT) is commonly used for mangoes to control fruit fly infestation and surface pathogens.

QUESTION 108

Which of the following is a non-destructive method of assessing fruit maturity?

- a. Penetrometer test
- b. Near-Infrared (NIR) spectroscopy
- c. Refractometer use
- d. Manual squeezing

ANSWER



b. Near-Infrared (NIR) spectroscopy

Explanation: NIR spectroscopy allows internal quality evaluation without damaging the fruit.

QUESTION 109

The term “Modified Atmosphere Packaging” (MAP) refers to:

- a. Packaging under vacuum
- b. Altering the internal gas composition of the packaging
- c. Packaging with antioxidants
- d. Freezing produce before wrapping

ANSWER



b. Altering the internal gas composition of the packaging

Explanation: MAP maintains produce quality by reducing respiration rates using specific gas mixtures inside sealed packs.

QUESTION 110

Which postharvest treatment involves slowing down ripening and microbial growth using cold storage?

- a. Vacuum cooling
- b. Forced-air drying
- c. Refrigeration
- d. Blanching

ANSWER



c. Refrigeration

Explanation: Refrigeration is the most common method to slow down biological processes and extend the shelf life of perishables.



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